

SIMATS ENGINEERING
ASSESSMENT TOOL 3 – VIVA QUESTIONS

Course:	Code:	CO: CO5
Duration:	Total Marks: 100	Reg No / Name :

COURSE OUTCOME COVERED

CO5: *Analyze vulnerabilities in Windows and Linux systems and evaluate security weaknesses in messaging, web, and mobile applications using appropriate exploitation techniques and hacking tools. (BT5)*

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Instructions: Answer all questions. Each question carries 4 marks.

1. How would you evaluate the security posture of a Windows system after identifying multiple open services?
2. Which factors would you consider when choosing a hacking tool for Windows vulnerability assessment, and why?
3. How would you evaluate the security risk associated with an exposed RPC service?
4. Why are Windows Null Sessions considered a security weakness, and how would you assess their impact?
5. A Windows shared folder allows unauthorized users to access sensitive files. How would you evaluate the severity of this vulnerability?
6. How would you distinguish between a vulnerable Windows service and a misconfigured Windows service during security testing?
7. How would you evaluate the effectiveness of information-gathering techniques against a Windows target?
8. If a Windows vulnerability can be exploited but requires local access, how would you classify its security impact compared with a remotely exploitable vulnerability?
9. How would you evaluate the security risks of unconnected or unnecessary services running on a Linux system?

10. Why are rhosts and hosts.equiv considered security risks in Linux? How would you evaluate their impact on system security?
11. How would you assess whether an NFS configuration exposes sensitive Linux files to unauthorized users?
12. A Linux file has incorrect permissions that allow unauthorized modification. How would you evaluate the resulting security risk?
13. How would you compare the security impact of weak file permissions with an unpatched Linux vulnerability?
14. How would you evaluate whether a buffer overflow vulnerability in a Linux application is practically exploitable?
15. Which factors would you consider before selecting an exploitation technique for a Linux vulnerability?
16. How would you evaluate the security of a messaging application that stores user credentials or messages insecurely?
17. A web application is vulnerable to SQL injection. How would you evaluate the severity of this vulnerability and its possible consequences?
18. How would you determine whether a web application vulnerability represents a critical security risk or a low-risk finding?
19. How would you evaluate the security of a mobile application using a tool such as MobSF?
20. A mobile application requests unnecessary permissions such as contacts, microphone, and location. How would you evaluate the associated security risk?
21. How would you evaluate the risk of hardcoded API keys or passwords discovered during mobile application analysis?
22. If the same vulnerability is found in both a web application and a mobile application, how would you compare their security impact?
23. How would you select appropriate hacking tools for assessing Windows, Linux, web, and mobile applications?
24. After performing vulnerability exploitation, how would you evaluate whether the exploitation technique actually demonstrates a significant security weakness?
25. You have identified vulnerabilities in Windows, Linux, messaging, web, and mobile applications. How would you prioritize them and prepare a professional security report?

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Mark
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Understanding of Problem Context	20	Clearly understands the security problem and accurately identifies vulnerabilities, risks, and requirements	Good understanding of the security problem with minor gaps	Basic understanding; some security requirements or vulnerabilities are unclear	Poor understanding or misinterpretation of the security problem	
Application of Concepts	30	Accurately applies appropriate ethical hacking concepts, tools, and exploitation techniques to analyze complex security problems	Applies relevant security concepts and tools correctly with minor errors	Applies basic concepts with limited accuracy or tool selection	Fails to apply appropriate security concepts or tools	
Problem Solving Approach	20	Demonstrates a logical, well structured, systematic, and efficient approach to identifying and evaluating security weaknesses	Mostly logical and systematic approach with minor gaps	Basic approach with some inconsistencies in vulnerability analysis	Illogical, unclear, or inappropriate approach to solving the security problem	

Accuracy of Solution	20	Provides a completely correct security analysis with appropriate justification, evidence, and suitable prevention	Mostly correct analysis with minor errors or gaps	Partially correct analysis with limited justification	Incorrect or incomplete security analysis	
Clarity	10	techniques, Provides clear, well-organized, technically sound, and properly explained responses using appropriate	Understandable response with minor clarity or terminology issues	Limited clarity and explanation; uses technical terms inconsistently	Poorly presented, unclear, and difficult to understand	

		security				
Total	100	terminology				/100

1. How would you evaluate the security posture of a Windows system after identifying multiple open services?

I would use **Nmap** to identify open ports, services, and versions. Then I would check the services for known **CVEs, outdated software, and weak authentication**. Necessary services should be kept, while unnecessary ones should be disabled. I would also check whether services are externally accessible or protected by a firewall. High-risk services such as old **SMB, Telnet, FTP, or RPC** would be prioritized. Finally, I would rate the system as **Critical/High/Medium/Low** and recommend patching, disabling unused services, and restricting network access.

3. How would you evaluate the security risk associated with an exposed RPC service?

I would check whether **RPC port 135** and related dynamic ports are exposed and identify the RPC services running. The risk depends on **network exposure, authentication, patch status, and known vulnerabilities**. An unpatched RPC service exposed to the Internet may allow **remote code execution or information disclosure** and would be rated critical. A patched, authenticated, and internally restricted RPC service would have lower risk.

7. How would you evaluate the effectiveness of information-gathering techniques against a Windows target?

I would evaluate techniques based on **coverage, accuracy, usefulness, stealth, and time**. Passive methods such as **OSINT, DNS, and WHOIS** can reveal basic information, while active methods such as port scanning and SMB enumeration reveal services, users, and shares. The most effective techniques provide **accurate and actionable information** with minimum noise and detection.

19. How would you evaluate the security of a mobile application using a tool such as MobSF?

Using **MobSF**, I would perform static and dynamic analysis. Static analysis checks **permissions, exported components, hardcoded secrets, weak cryptography, and vulnerable libraries**. Dynamic analysis checks **network traffic, TLS protection, storage, and API activity**. I would verify MobSF findings manually and prioritize them based on **impact and exploitability**, using the **OWASP Mobile Top 10** for reporting.

20. A mobile application requests unnecessary permissions such as contacts, microphone, and location. How would you evaluate the associated security risk?

I would compare requested permissions with the application's actual functionality using the **principle of least privilege**. Permissions such as **microphone, location, and contacts** can expose sensitive information. I would check whether these permissions are actually used and whether collected data is transmitted or

disclosed in the privacy policy. Unnecessary permissions that access or transmit sensitive data would be considered **high risk** because they increase privacy and security exposure.